1. Clinical Question & Core Hypothesis

Clinical Question

Intratumoral TLS-like lymphoid aggregates are expected to enhance local antitumor immunity, yet some TLS-rich tumors fail to respond to immunotherapy (here represented by BCG); the provided path links oncogenic drivers in stem-like tumor cells to PD-L1 upregulation, suggesting a tumor-intrinsic mechanism that could disable TLS function. 2) Tier 1 path(s) used: the single path connecting TMPRSS2/ERG expression in stem-like (PSMA+) cancer cells to PI3K-driven upregulation of PD-L1 (CD274). I built on TMPRSS2/ERG (oncogenic driver in the path) and CD274 (PD-L1, the immune checkpoint effector) because they form the most direct mechanistic axis that could neutralize TLS activity. 3) Tier 2 supporting evidence: PSMA expression on cancer stem cells (links stemness/therapy resistance to the axis) and PI3K pathway activation (mechanistic mediator of CD274 induction) reinforce the core TMPRSS2/ERG→PD-L1 hypothesis and suggest a cell-population (PSMA+ stem cells) and signaling mechanism (PI3K) for intervention. 4) Why therapy_response as the endpoint: the path explicitly connects PD-L1 modulation to BCG (an immunotherapy) and the clinical question is whether TLS-rich tumors still benefit from intravesical immunotherapy; 'therapy_response' is the most direct, clinically actionable endpoint. 5) Deliberate exclusions: I excluded broader carcinoma-level edges and general organ-level context (bladder/urothelial descriptors) to keep the hypothesis mechanistically focused; PSMA and PI3K are moved to secondary_modifiers rather than being placed in the core hypothesis to respect the 1–2 anchor-gene limit?

Working Hypothesis

In tumors with abundant TLS-like lymphoid aggregates, TMPRSS2-ERG expression in stem-like cancer cells drives upregulation of CD274 (PD-L1) via PI3K pathway signaling, functionally suppressing TLS-mediated antitumor immunity. Consequently, TLS-rich tumors with high TMPRSS2-ERG–CD274 activity will show reduced response to BCG therapy, identifying an immune-resistant TLS-rich subset.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TMPRSS2/ERG, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMPRSS2/ERG: TMPRSS2/ERG (oncogenic driver in the path) and CD274 (PD-L1, the immune checkpoint effector) because they form the most direct mechanistic axis that could neutralize TLS activity.

CD274: CD274 induction) reinforce the core TMPRSS2/ERG→PD-L1 hypothesis and suggest a cell-population (PSMA+ stem cells) and signaling mechanism (PI3K) for intervention.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PSMA (FOLH1)	Supporting modifier from knowledge graph
Secondary	PI3K pathway (e.g., PIK3CA)	Supporting modifier from knowledge graph
Secondary	cancer stem cell phenotype	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PSMA (FOLH1): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PI3K pathway (e.g., PIK3CA): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.